CASE 7
OMENN SYNDROME

A defect in V(D)J recombination results in severe immunodeficiency.

The development of B cells in the bone marrow and T cells in the thymus is initiated by the assembly of Variable (V), Diversity (D), and Joining (J) segments of the immunoglobulin and the T cell receptor (TCR) genes, respectively (Fig. 7.1). This process is called V(D)J recombination. In the case of the immunoglobulin light chain and the TCR α chain, V and J segments are joined together. The immunoglobulin heavy chain locus and the TCR β locus require an additional component, a D element, which is located between the V and J gene segments. First, a D and a J gene segment are joined, and then a V gene segment is added to form the rearranged VDJ exon. In all of these recombination events, the DNA between the rearranging gene segments is deleted from the chromosome. Because there are many different V, D, and J segments in the germline genome, there are several million possible combinations. This rearrangement mechanism is how much of the vast diversity in the antibody and T-cell receptor repertoires is generated. Moreover, small insertions or deletions of nucleotides at the junctions between V and D, and D and J segments further contribute to diversity.

The process of V(D)J recombination is initiated by enzymes encoded by the recombinase-activating genes *RAG1* and *RAG2*. The RAG1 and RAG2 enzymes form a tetrameric complex (with two subunits each of RAG1 and RAG2) capable of nicking double-stranded DNA. They recognize canonical DNA sequences that flank the coding gene segments. Each recombination signal sequence consists of a

TOPICS BEARING ON THIS CASE:

- V(D)J recombination
- RAG enzymes
- Severe combined immunodeficiency

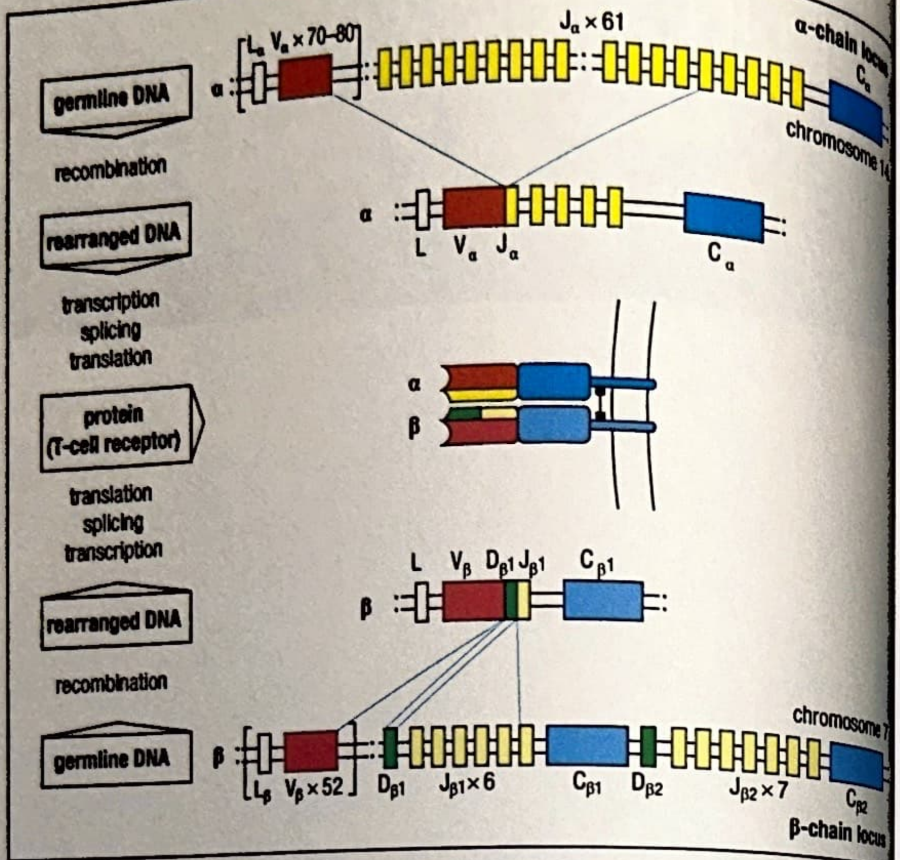

Fig. 7.1 Rearrangement of the T-cell receptor genes. The top and bottom rows of the figure show the germline arrangement of the variable (V), diversity (D), joining (J), and constant (C) gene segments at the T-cell receptor α and β loci, respectively. During T-cell development, a V-region sequence for each chain is assembled by DNA recombination. For the α chain (top), a $V_α$ gene segment rearranges so that it is next to a $J_α$ gene segment, creating a functional gene encoding the V domain. For the β chain (bottom), rearrangement of a $D_β$, a $J_β$, and a $V_β$ gene segment creates the functional V-domain exon. A similar array of gene segments is present at the immunoglobulin loci, and immunoglobulin gene rearrangement follows an essentially similar course.

heptamer (CACAGTG) followed by a spacer of 12 or 23 bases and then a nonam (ACAAAAACC) (Fig. 7.2). RAG1 binds to the nonamer element; subsequen RAG2 stabilizes occupancy of the heptamer by RAG1. The DNA sequence that for the border between the heptamer and the coding segment is then nicked, an break in the double-stranded DNA occurs. The coding ends are initially sealed b hairpin. Next, a series of ubiquitously expressed proteins (Ku70, Ku80, DNA-PK Artemis, DNA ligase IV (LIG4), XRCC4, and Cernunnos/XLF) are recruited t mediate DNA repair and rejoining of coding and signal ends (Fig. 7.3).

If either of the *RAG* genes is knocked out in mice, the development of B cells a T cells is completely abolished and the mice have severe combined immunode ciency. Null mutations in *RAG1* and *RAG2* have also been found in human SC cases that lack both T and B cells, but have NK cells ($T^-B^-NK^+$ SCID). In additi defects of Artemis, LIG4, and DNA-PK have also been identified in patients w $T^-B^-NK^+$ SCID; and mutations of Cernunnos/XLF cause combined immun deficiency with markedly reduced numbers of T and B lymphocytes. Howev hypomorphic mutations in *RAG* (and less often also in *Artemis*, *LIG4*, or oth

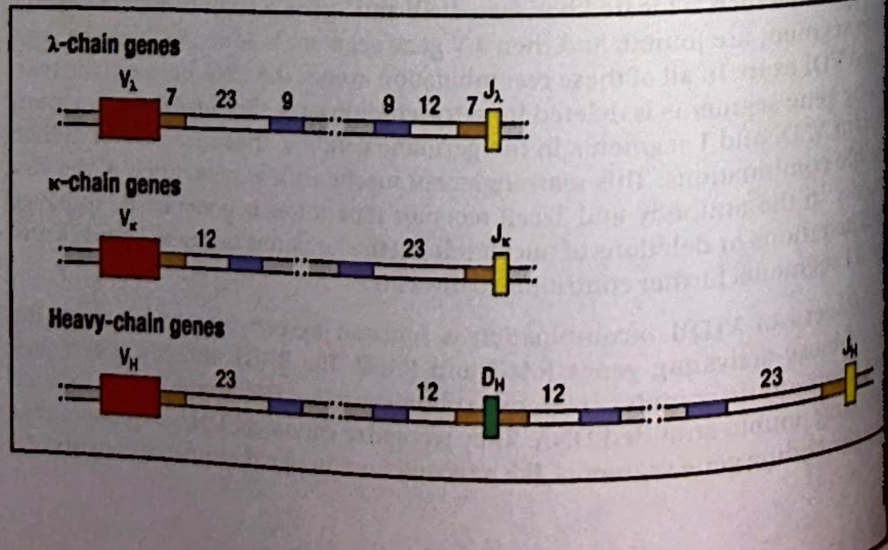

Fig. 7.2 Each V, D, or J gene segment is flanked by recombination signal sequences (RSSs). The immunoglobulin genes and their RSSs are shown here. There are two types of RSS. One consists of a nonamer (9 nucleotides, shown in purple) and a heptamer (7 nucleotides, shown in orange) separated by a spacer of 12 nucleotides (white). The other consists of the same 9- and 7-nucleotide sequences separated by a 23-nucleotide spacer (white).

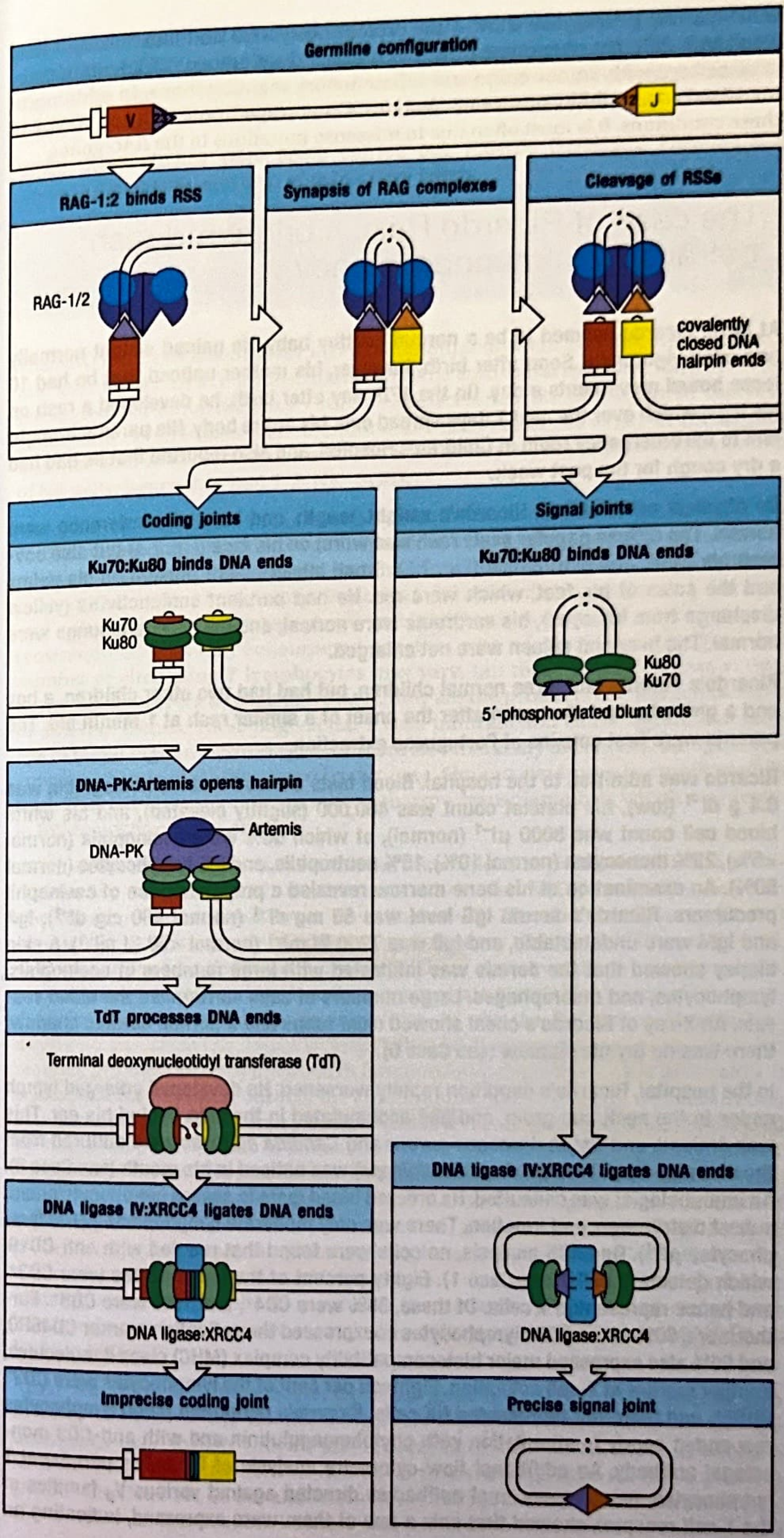

Fig. 7.3 Steps in the V(D)J recombination process. V(D)J recombination is initiated by the lymphocyte-specific RAG-1 and RAG-2 enzymes, which recognize the recombination signal sequences (RSSs) that flank the coding variable (V), diversity (D), and joining (J) elements. The DNA double-strand break at the coding ends is initially sealed by a hairpin. In the second step of the process, the ubiquitously expressed Ku70 and Ku80 are recruited both at coding ends and at signal ends. DNA-protein kinase catalytic subunit (DNA-PKcs) and Artemis are also recruited to the coding ends, and Artemis mediates opening of the coding-end hairpins. The enzyme terminal nucleotide transferase (TdT) may introduce additional nucleotides at the junction between coding elements. Finally, the enzymes DNA ligase IV and XRCC4 (involved in DNA repair and ligation) are recruited at both coding and signal ends and mediate the formation of coding and signal joints. Another enzyme, Cernunnos/XLF (not shown in the figure), also participates in the DNA repair process.

Fig. 7.4 **Bright scaly red rash on the face and shoulders of an infant with Omenn syndrome.**

One-month-old infant with bright red rash and purulent conjunctivitis. Admit to hospital.

Eosinophilia and low lymphocyte count. No thymic shadow.

Fig. 7.5 **Legs and groin of an infant with Omenn syndrome.** The skin is bright red and wrinkled from edema and the infiltration of inflammatory cells.

SCID-causing genes) may allow some protein expression and function, and may result in a different phenotype in which residual development of T lymphocytes is associated with autoimmune and inflammatory manifestations, in addition to increased susceptibility to severe infections. Omenn syndrome is the prototype of these conditions. It is most often due to missense mutations in the *RAG* genes.

The case of Ricardo Reis: a bright red rash betrays an immunodeficiency.

At birth, Ricardo seemed to be a normal healthy baby. He gained weight normally and cried vigorously. Soon after birth, however, his mother noticed that he had 10 loose bowel movements a day. On the 17th day after birth, he developed a rash on his legs, which over the next 7 days spread over his entire body. His parents brought him to the emergency room at Children's Hospital, and also reported that he had had a dry cough for the past week.

On physical examination, Ricardo's weight, length, and head circumference were normal. The diffuse papular scaly rash was worst on his face (Fig. 7.4) but also covered his trunk and extremities (Fig. 7.5). Small blisters were present on his palms and the soles of his feet, which were red. He had purulent conjunctivitis (yellow discharge from his eyes), his eardrums were normal, and his heart and lungs were normal. The liver and spleen were not enlarged.

Ricardo's parents had three normal children, but had had two other children, a boy and a girl, who had died soon after the onset of a similar rash at 1 month old. The parents were first cousins of Portuguese extraction.

Ricardo was admitted to the hospital. Blood tests showed that his hemoglobin was 8.4 g dl^{-1} (low), his platelet count was 460,000 (slightly elevated), and his white blood cell count was 8000 μl^{-1} (normal), of which 56% were eosinophils (normal <5%), 23% monocytes (normal 10%), 15% neutrophils, and 6% lymphocytes (normal 50%). An examination of his bone marrow revealed a preponderance of eosinophil precursors. Ricardo's serum IgG level was 55 mg dl^{-1} (normal 400 mg dl^{-1}), IgA and IgM were undetectable, and IgE was 7200 IU ml^{-1} (normal <50 IU ml^{-1}). A skin biopsy showed that the dermis was infiltrated with large numbers of eosinophils, lymphocytes, and macrophages. Large numbers of cells surrounded the blood vessels. An X-ray of Ricardo's chest showed clear lungs and a normal cardiac shadow; there was no thymic shadow (see Case 6).

In the hospital, Ricardo's condition rapidly worsened. He developed enlarged lymph nodes in the neck and groin, and pus accumulated in the skin behind his ear. This was drained, and *Staphylococcus aureus* and *Candida albicans* were cultured from the drainage fluid. Thrush (*Candida albicans*) was noticed in his mouth (see Case 5). An immunologist was consulted. He ordered blood tests to assess lymphocyte count, subset distribution, and function. There was only moderate lymphopenia (2100 lymphocytes μl^{-1}). On FACS analysis, no cells were found that reacted with anti-CD19, which detects B cells (see Case 1). Eighty percent of the lymphocytes were $CD3^+$, and hence represented T cells. Of these, 85% were $CD4^+$, and 15% were $CD8^+$. Furthermore, 90% of the $CD3^+$ lymphocytes coexpressed the activation marker CD45RO, and 65% also expressed major histocompatibility complex (MHC) class II molecules, another marker of T-cell activation. Eighteen per cent of the lymphocytes were $CD3^-$ $CD56^+$, and therefore represented NK cells. Ricardo's peripheral blood lymphocytes responded poorly to stimulation with phytohemagglutinin and with anti-CD3 monoclonal antibody. An additional flow-cytometry analysis of Ricardo's peripheral T lymphocytes, using monoclonal antibodies directed against various V_β families of the T-cell receptor, showed that only a few of them were expressed, indicating an

oligoclonal T-cell receptor repertoire. The *RAG1* and *RAG2* genes were sequenced, and homozygosity for the Arg229Gln (R229Q) mutation was found in the *RAG2* gene. The T cells were definitively identified as Ricardo's (and not as transferred maternal T cells) by HLA typing. Ricardo was diagnosed with Omenn syndrome.

While these studies were being carried out, Ricardo developed *Pneumocystis jirovecii* pneumonia and died of respiratory failure.

Lymph nodes enlarged.
Opportunistic infections noted.
Immunodeficiency?

Omenn syndrome.

The RAG enzymes essential for V(D)J recombination were first discovered in mice and later identified in humans. Infants with the autosomal recessive form of severe combined immunodeficiency (SCID) were screened for mutations in the *RAG* genes, and several cases were identified in which either the *RAG1* or *RAG2* gene was mutated. These infants lacked T and B lymphocytes, but had a normal number of NK cells; hence they had $T^-B^-NK^+$ SCID.

The majority of patients with Omenn syndrome have missense mutations in the *RAG* genes resulting in reduced gene function. The result is that enzyme activity is profoundly decreased, but not absent. Omenn syndrome is characterized by early-onset generalized skin rash (erythroderma), failure to thrive, protracted diarrhea, and enlargement of the liver, spleen, and lymph nodes. A high eosinophil count (eosinophilia) is usually encountered, together with a lack of B lymphocytes. The number of circulating T lymphocytes may vary, but these T cells express activation markers and are oligoclonal; that is, they are the products of a limited number of different clones. These oligoclonal T cells infiltrate and cause significant damage in target organs. Immunoglobulins are also markedly decreased, but IgE levels are typically raised. As illustrated by this case, Omenn syndrome is usually rapidly fatal. The only known treatment is bone marrow transplantation, which may result in full correction of the disease.

Genetic defects that result in a severe, but incomplete, impairment of T-cell development by interfering with mechanisms other than V(D)J recombination can also result in Omenn syndrome. These include IL-7Rα chain deficiency (IL-7 is required for lymphocyte development), γ_c deficiency (X-linked SCID; see Case 5), and mutations of the *RMRP* gene. The last of these causes cartilage hair hypoplasia, a condition characterized by dwarfism, sparse hair, a variable degree of immunodeficiency, and hematological abnormalities. As with the *RAG* genes, Omenn syndrome occurs when the defect is 'leaky'; that is, due to a missense mutation that severely impairs but does not abolish function, allowing a few T cells to develop. It is likely that in Omenn syndrome the autoimmune manifestations, with infiltration of target organs by oligoclonal T cells, reflect several mechanisms, as demonstrated by studies in patients and in animal models of the disease. Poor generation of T lymphocytes in the thymus results in impaired maturation of medullary thymic epithelial cells and reduced expression of *AIRE*, thus impinging on the deletion of self-reactive T cells (see Case 17). Furthermore, generation of regulatory T cells in the thymus is also impaired, affecting peripheral tolerance (see Case 18). Finally, the few T cells that are generated in the thymus of patients with Omenn syndrome undergo extensive peripheral expansion (homeostatic proliferation) and secrete increased amounts of cytokines, including inflammatory (IFN-γ) and TH2 (IL-4, IL-5) cytokines.

Apart from the lymphocyte-specific RAG proteins, V(D)J recombination also involves proteins of the nonhomologous end-joining (NHEJ) pathway that are universally used for DNA repair and recombination in human cells. In addition to $T^-B^-NK^+$ SCID, patients with defects in genes that encode for these proteins (ARTEMIS, DNA-PK, LIG4, and Cernunnos/XLF) present increased cellular sensitivity to ionizing radiation, because they are unable to repair radiation-induced

DNA damage. These radiosensitive forms of $T^-B^-NK^+$ SCID are often associated with extraimmune clinical manifestations, such as microcephaly, neurodevelopmental problems, and growth and development defects. For patients with defects of NHEJ, use of radiomimetic drugs in the conditioning regimen of bone marrow transplantation should be avoided or kept to a minimum in order to avoid serious adverse effects.

Questions.

1 *How do you explain the high IgE level and eosinophilia in this patient?*

2 *How do you explain the enlargement of the lymph nodes in this patient?*

3 *How does Ricardo's family history help you determine the mode of inheritance of Omenn syndrome?*

4 *A bright red rash (erythroderma) is characteristic of Omenn syndrome. What causes this rash?*

CASE 8
MHC CLASS II DEFICIENCY

An inherited failure of gene regulation.

The class II molecules of the major histocompatibility complex (MHC) are involved in presenting antigens to $CD4^+$ T cells. The peptide antigens that they present are derived from extracellular pathogens and proteins taken up into intracellular vesicles, or from pathogens such as *Mycobacterium* that persist intracellularly inside vesicles. MHC class II molecules are expressed constitutively on antigen-presenting cells, including dendritic cells, monocyte/macrophages, and B lymphocytes. In humans, MHC class II molecules, together with MHC class I molecules (see Case 12), are known as the HLA antigens. They are also expressed on the epithelial cells of the thymus and their expression can be induced on other cells, principally by the cytokine interferon-γ. T cells also express MHC class II molecules when they are activated.

MHC class II molecules are heterodimers consisting of an α chain and a β chain (Fig. 8.1). The genes encoding both chains are located in the Human Leukocyte Antigen (HLA) locus on the short arm of chromosome 6 in humans (Fig. 8.2). The principal MHC class II molecules are designated DP, DQ, and DR, and, like the MHC class I molecules, they are highly polymorphic. Peptides bound to MHC class II molecules can be recognized only by the T-cell receptors of $CD4^+$ T cells and not by those of $CD8^+$ T cells (Fig. 8.3). MHC class II molecules expressed in the thymus also have a vital role in the intrathymic maturation of $CD4^+$ T cells.

Expression of the genes encoding the α and β chains of MHC class II molecules must be strictly coordinated and it is under complex regulatory control by a series of transcription factors. The existence of these transcription factors and a means of identifying them were first suggested by the study of patients with MHC class II deficiency.

TOPICS BEARING ON THIS CASE:

- Role of MHC class II molecules in antigen presentation to CD4 T cells
- Role of co-receptor molecule CD4 in antigen recognition by T cells
- Intrathymic maturation of CD4 T cells
- Mixed lymphocyte reaction
- Lymphocyte stimulation by polyclonal mitogens
- FACS analysis